

PROJECT REPORT ON

Ride Sharing Driver Performance Analysis

using PySpark

SUBMITTED BY

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1. Introduction

The **Ride Sharing Driver Performance Analysis** project is an end-to-end **Data Engineering** project developed using **PySpark**. The primary objective of this project is to process raw ride-sharing datasets, perform data cleaning and transformation, and generate meaningful business insights that can help ride-sharing companies improve their operations and decision-making.

In today's digital world, ride-sharing platforms such as **Uber, Ola, and Lyft** generate a huge amount of data every day. This data includes information about drivers, trips, pickup and drop locations, fares, trip durations, customer ratings, and cancellations. Processing such a large amount of data manually is difficult and time-consuming. Therefore, Data Engineering techniques are used to organize, clean, and transform the raw data into a structured format that can be easily analyzed.

This project follows the **Medallion Architecture**, which is a modern data engineering design pattern consisting of three layers: **Bronze, Silver, and Gold**. The Bronze layer stores the raw data in its original form, the Silver layer contains cleaned and transformed data, and the Gold layer provides business-ready datasets that can be directly used for reporting and analysis. This layered approach improves data quality, simplifies data management, and makes the overall pipeline more efficient and scalable.

The project uses three datasets: **Drivers, Trips, and Trip Logs**. The Drivers dataset contains driver details such as driver ID, name, city, and rating. The Trips dataset stores ride-related information such as pickup location, drop location, distance, fare amount, and trip status. The Trip Logs dataset contains operational details such as trip start time, end time, delay minutes, and cancellation information. These datasets are integrated using PySpark to create a single unified dataset for further analysis.

Several data engineering operations are performed during the project, including **data ingestion, data quality checks, dataset joining, creation of derived columns, filtering invalid records, and storing data in different Medallion layers using the Parquet file format**. Finally, multiple business Key Performance Indicators (KPIs) are generated, such as **driver performance analysis, cancellation rate analysis, revenue analysis, pickup location analysis, and driver ranking using Spark Window Functions**.

Overall, this project demonstrates the practical implementation of an ETL (Extract, Transform, Load) pipeline using PySpark and provides a strong foundation in modern Data Engineering concepts. It also highlights how raw operational data can be transformed into valuable business information that supports better decision-making and improves the efficiency of ride-sharing services.

Data Engineering plays a significant role in modern organizations because business decisions are based on accurate and reliable data. Before data can be analyzed, it must be collected, cleaned, transformed, and stored in a structured format. This project demonstrates the complete lifecycle of data processing, starting from raw CSV files and ending with business-ready datasets. It provides practical exposure to the ETL (Extract, Transform, Load) process, which is one of the most important concepts in Data Engineering.

2. Objective

The primary objective of this project is to design and implement a complete **Data Engineering pipeline** using **PySpark** by following the **Medallion Architecture (Bronze, Silver, and Gold)**. The project aims to demonstrate how raw ride-sharing data can be collected, processed, cleaned, transformed, and analyzed to generate meaningful business insights. Through this project, the complete lifecycle of data processing is implemented, starting from raw CSV files and ending with business-ready analytical datasets.

Another important objective of this project is to gain practical exposure to **ETL (Extract, Transform, Load)** concepts that are widely used in modern Data Engineering. The project focuses on handling structured datasets efficiently, improving data quality, and preparing reliable data for business analysis. It also helps in understanding how organizations use data pipelines to convert raw operational data into valuable information for decision-making.

The project is designed not only to perform technical operations but also to solve real-world business problems in the ride-sharing domain. By analyzing driver performance, trip information, revenue, cancellations, and customer demand, the project provides useful insights that can help improve operational efficiency and customer satisfaction.

The main objectives of this project are:

- To read and process multiple CSV datasets using PySpark.
- To implement the Medallion Architecture (Bronze, Silver, and Gold).
- To perform data cleaning and quality checks.
- To join multiple datasets into a single dataset.
- To generate business insights such as driver performance, revenue, and cancellation rate.
- To understand the basic concepts of Data Engineering and ETL pipelines.
- To generate business insights such as **driver performance, revenue analysis, cancellation rate, pickup location analysis, and driver ranking**.
- To understand the importance of **Parquet format** for faster storage and improved query performance.
- To gain hands-on experience with **ETL (Extract, Transform, Load) pipeline development** using PySpark.
- To understand how Data Engineering pipelines help organizations convert raw data into meaningful business information.
- To build a beginner-friendly project that demonstrates practical implementation of modern Data Engineering concepts and can be used for learning, academic submission, and interview preparation.
- To apply **data quality checks** to improve the accuracy and reliability of the datasets.
- To transform raw data into a clean and analytics-ready dataset in the Silver layer.

3. Tools and Technologies Used

The successful implementation of this project required the use of several tools and technologies, each serving a specific purpose throughout the data engineering pipeline. These technologies were selected to efficiently perform data ingestion, processing, transformation, storage, and analysis while following modern Data Engineering practices. PySpark and Apache Spark were used as the core processing framework, whereas Python provided the programming environment for implementing the ETL pipeline. Visual Studio Code and Jupyter Notebook were used for development and execution of the project, while the Parquet file format was used for efficient storage of processed datasets in the Bronze, Silver, and Gold layers. Together, these tools enabled the development of a scalable, organized, and industry-inspired data processing workflow.

- Python 3.12
- PySpark 3.5.1
- Apache Spark
- Jupyter Notebook
- Visual Studio Code (VS Code)
- Parquet File Format

4. Dataset Used

The project uses three different datasets that together provide complete information about the ride-sharing system. Each dataset represents a specific aspect of the business, including driver information, trip details, and trip execution logs. These datasets are interconnected through common keys such as Driver ID and Trip ID, allowing them to be joined and analyzed as a single unified dataset. The combination of these datasets enables the generation of meaningful business insights such as driver performance, cancellation analysis, revenue generation, and demand patterns. Before analysis, all datasets were loaded into PySpark DataFrames and processed through the Bronze, Silver, and Gold layers of the Medallion Architecture.

4.1 Drivers Dataset

The Drivers Dataset contains information related to the drivers registered on the ride-sharing platform. It stores personal and operational details of each driver, which are later used to evaluate driver performance and analyze city-wise operations.

Contains driver details such as:

- Driver ID
- Driver Name
- City
- Rating

4.2 Trips Dataset

The Trips Dataset stores the details of every ride booked on the ride-sharing platform. It contains trip-related information such as pickup location, drop location, distance traveled, fare amount, and trip status. This dataset forms the core of the project because it represents the actual business transactions performed by drivers.

Contains trip information such as:

- Trip ID
- Driver ID
- Pickup Location
- Drop Location
- Distance
- Fare Amount
- Trip Status

4.3 Trip Logs Dataset

The Trip Logs Dataset contains operational information generated during the execution of each trip. It stores timestamps, delays, and cancellation details that help analyze ride completion and operational efficiency.

Contains trip execution details such as:

- Start Time
- End Time
- Delay Minutes
- Cancellation Flag

5. Project Workflow

The project follows a structured Data Engineering workflow to process raw ride-sharing datasets into meaningful business insights. Instead of directly analyzing the raw data, the project processes the data through multiple stages such as data ingestion, storage, cleaning, transformation, validation, and analytics. This structured workflow ensures that the data remains organized, accurate, and suitable for business reporting.

The entire pipeline is designed according to the Medallion Architecture, where data flows through three different layers: Bronze, Silver, and Gold. Each layer has a specific responsibility in the data processing pipeline. The Bronze layer stores the raw data exactly as received, the Silver layer contains cleaned and transformed data, and the Gold layer stores business-ready analytical datasets. This layered approach improves data quality, simplifies maintenance, and makes the pipeline scalable for future enhancements.

During the implementation of this project, multiple Data Engineering operations were performed, including reading CSV datasets using PySpark, storing raw data in Parquet format, performing data quality checks, joining multiple datasets, creating derived columns, filtering invalid records, generating business KPIs, and applying Spark optimization techniques. Finally, the processed datasets were stored in the Gold layer to support business analysis and reporting.

The workflow was executed in a step-by-step manner to ensure that each stage of the pipeline produced reliable and validated data before moving to the next stage. This

approach closely resembles real-world Data Engineering pipelines used by organizations for processing large volumes of operational data. The complete workflow implemented in this project is illustrated through the following steps.

Step 1 – Data Ingestion

The CSV datasets are loaded into PySpark DataFrames.

Step 2 – Bronze Layer

The raw CSV files are stored in Parquet format without making any changes. This layer keeps the original data safely for future use.

Step 3 – Data Cleaning

Several data quality checks are performed:

- Null value checking
- Duplicate record checking
- Invalid distance checking
- Invalid rating checking

Step 4 – Data Transformation

The datasets are joined using Driver ID and Trip ID. New columns such as Trip Duration and Completion Flag are created for analysis.

Step 5 – Silver Layer

The cleaned and transformed dataset is stored in the Silver layer.

Step 6 – Gold Layer

Business reports are generated from the Silver layer, including:

- Driver Performance
- Cancellation Rate
- Revenue Analysis
- Pickup Location Analysis
- Driver Ranking

6. Results

The implementation of the Ride Sharing Driver Performance Analysis project was successfully completed using PySpark by following the Medallion Architecture (Bronze, Silver, and Gold). The project successfully processed the raw ride-sharing datasets, performed data cleaning and transformations, and generated meaningful business insights from the available data.

Initially, all three datasets (Drivers, Trips, and Trip Logs) were successfully loaded into PySpark DataFrames and stored in the Bronze Layer without any modifications. This ensured that the original data was preserved for future reference. The datasets were then

validated through multiple data quality checks to identify null values, duplicate records, invalid ratings, negative distances, and incorrect records.

After completing the validation process, the datasets were joined using common keys such as Driver ID and Trip ID. Additional derived columns, including Trip Duration and Completion Flag, were created to improve the analytical capabilities of the dataset. The cleaned and transformed data was then stored in the Silver Layer using the Parquet file format.

In the final stage of the project, several business-oriented analytical reports were generated and stored in the Gold Layer. These reports provide valuable insights that can help ride-sharing companies monitor operations, evaluate driver performance, and support business decision-making.

The project successfully generated several business insights:

- Total trips completed by each driver.
- Average driver rating.
- Average fare collected by drivers.
- Driver cancellation percentage.
- Revenue generated by different cities.
- Most frequently used pickup locations.
- Driver ranking using Spark Window Functions.

7. Conclusion

This project successfully demonstrates the implementation of a complete **Data Engineering pipeline** using **PySpark** while following the **Medallion Architecture (Bronze, Silver, and Gold)**. The project provided practical experience in processing raw ride-sharing datasets and converting them into meaningful business information through a structured ETL (Extract, Transform, Load) workflow. By implementing different stages of data processing, the project helped in understanding how raw operational data can be transformed into clean, organized, and business-ready datasets.

Throughout the project, various Data Engineering concepts were implemented, including **data ingestion, data validation, data cleaning, dataset transformation, joins, aggregation, derived column creation, window functions, and business analytics**. The project also demonstrated the importance of maintaining data quality by identifying null values, duplicate records, invalid entries, and inconsistent data before performing analytical operations. These steps ensured that the final analytical reports were accurate and reliable.

The implementation of the **Medallion Architecture** played a significant role in organizing the data processing pipeline. The **Bronze Layer** preserved the raw data in its original form, the **Silver Layer** stored cleaned and transformed datasets, and the **Gold Layer** contained business-ready reports and KPIs. This layered approach improved the maintainability, scalability, and readability of the entire pipeline while following industry-inspired Data Engineering practices.

The use of the **Parquet file format** significantly improved storage efficiency by reducing storage space and enabling faster data retrieval compared to traditional CSV files. PySpark's

distributed processing capabilities and built-in transformation functions simplified data processing tasks such as filtering, joining, aggregating, and ranking records. Additionally, the implementation of **Spark caching** and **Window Functions** enhanced the overall performance of the pipeline and demonstrated practical optimization techniques used in modern data processing systems.

The project successfully generated several valuable business insights, including **driver performance metrics, cancellation rate analysis, revenue analysis, pickup location analysis, and driver ranking**. These analytical reports can help ride-sharing companies monitor operational performance, identify high-performing drivers, understand customer demand patterns, and support data-driven business decisions.

8. Future Scope

Although the Ride Sharing Driver Performance Analysis project successfully demonstrates the implementation of an end-to-end Data Engineering pipeline using PySpark and the Medallion Architecture, there are several opportunities to further enhance its capabilities. The current implementation is designed as a beginner-friendly project using locally available datasets. However, with additional technologies and larger datasets, the project can be extended into a more scalable and production-ready solution. One possible enhancement is the integration of real-time data streaming using technologies such as Apache Kafka or Apache Spark Structured Streaming. Instead of processing static CSV files, the pipeline can continuously process live ride-sharing data generated by drivers and customers. This would enable organizations to monitor ride activities, cancellations, and driver performance in real time.

The project can be enhanced by:

- Connecting to real-time ride-sharing data.
- Creating interactive dashboards using Power BI or Tableau.
- Storing data in cloud platforms such as Azure or AWS.